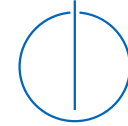


Robotics, Artificial Intelligence and Embedded Systems



Cognitive Systems

Part I: Foundations and Theory of Cognitive Systems

Chapter 3

Basic Concepts in Cognitive Science

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Topics in this Chapter

*Modeling the function...
... of the brain: from neuroanatomy to emergent systems*

- How is cognition reflected in the structure and function of the brain?
- What tools are available to observe cognition in the brain?
- Which formal tools are available to model cognitive processes?
- What is the difference between cognitivist and emergent systems?

Cognition in the Brain

The Beginnings of Cognitive Science

Phrenology

Phrenology, from grk. *phren* (mind) and *logos* (knowledge) was a predecessor of cognitive science founded by the German physician **Franz Joseph Gall** at the end of the 18th century. It tried to describe the mind based the on assumptions on the anatomical structure of the brain:

- The brain is the **organ of the mind**
- Human mental power is composed of **different independent faculties** that are located at specific regions of the brain
- The size of a brain region indicates how much the corresponding faculty contributes to a **person's character**
- There is a close correspondence between the sizes of brain regions and the shape of the skull

Adapted from Encyclopædia Britannica

Phrenology

Phrenology is **not a science**! The assignment of “**organs**” to brain regions is completely arbitrary. However, some of its main hypotheses are still valid in modern cognitive neuroscience:

- The brain as the center of thought
- **Modular organization** of the brain into anatomical regions
- Functional specialization of individual brain regions

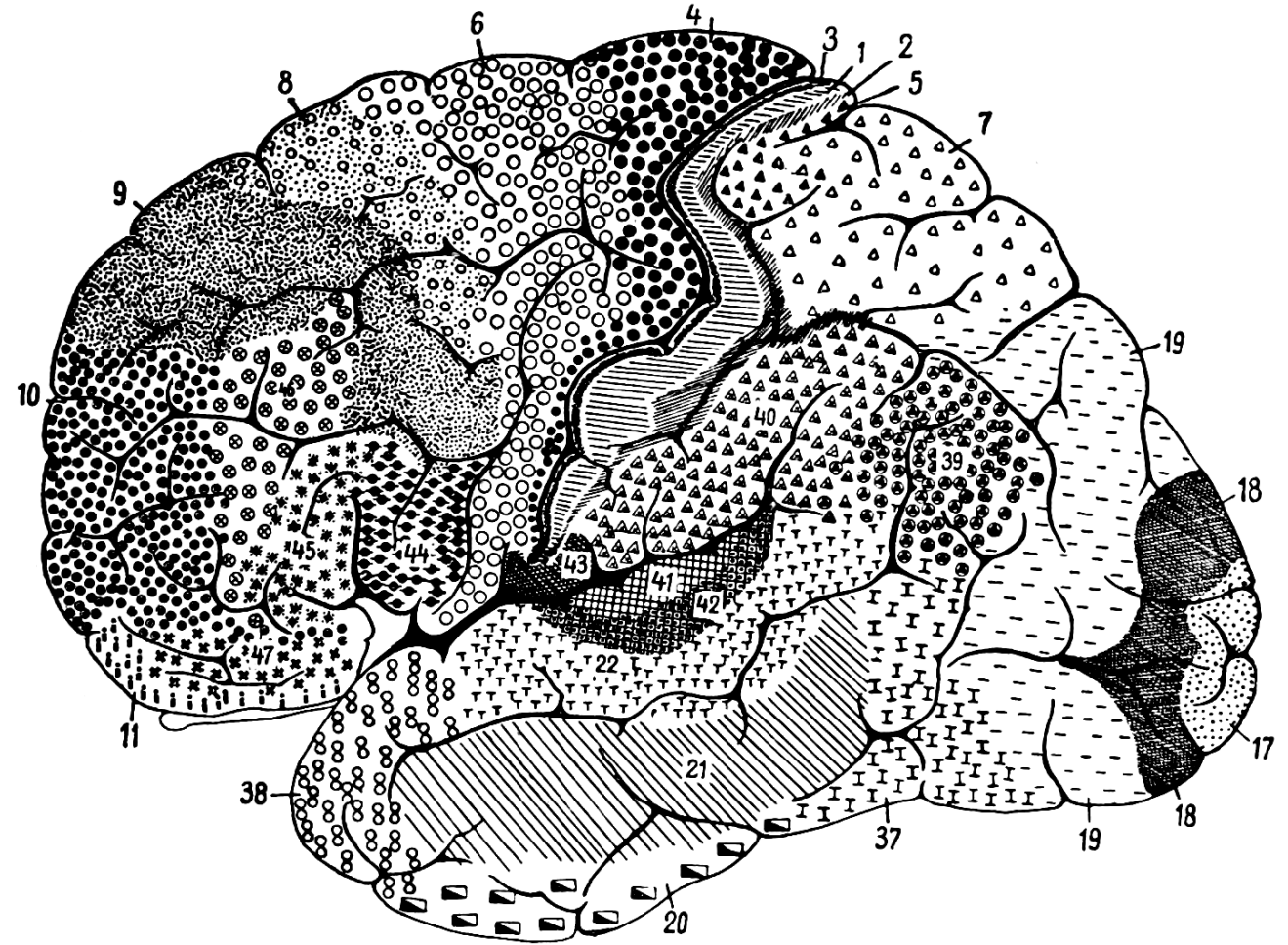


The Anatomical Architecture of the Human Brain

In 1907, the German neurologist Korbinian Brodman proposed to divide the cerebral cortex into 52 different regions based on **cytoarchitectonic properties**.

Today, these regions are known as the **Brodman areas**.

Brodman claimed that the individual regions correspond to **specific functions**.

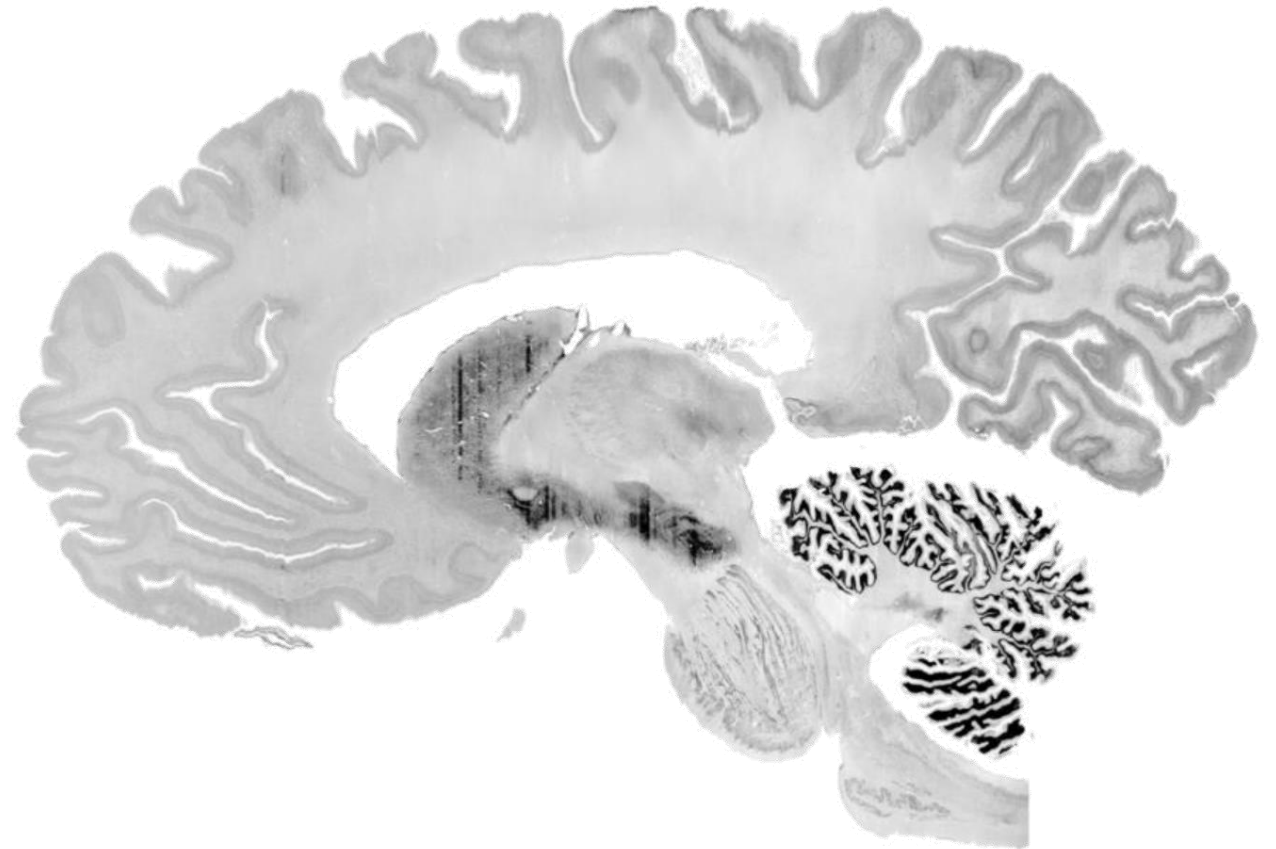


From K. Brodmann, Vergleichende Lokalisationslehre der Großhirnrinde: In ihren Prinzipien dargestellt auf Grund des Zellenbaues. Leipzig: Barth, 1909, p. 138.
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Modern Neuroanatomy

Ultrahigh-Resolution 3D Human Brain Model by Amunts et al. (2013)

- The brain is cut into **7400 histological sections** with thickness of 20 μm
- Every section is **stained** to identify the cell bodies
- The sections are **digitized** with a resolution of **13,000 by 11,000 pixels**
- Each section is processed to **remove artifacts** from the scanned images
- All slices are aligned for a **3D reconstruction**



Data from the Human Brain Project by Katrin Amunts and Alan C. Evans licensed under CC BY-NC-SA 4.0

Processing of Histological Brain Sections



Structural Neuroimaging

Magnetic Resonance Imaging (MRI)

MRI is a non-invasive method that is used for imaging the anatomic structure of tissue **in vivo**. It is an **active sensing** method based on the magnetic properties of hydrogen atoms that dissipate energy differently depending on the surrounding tissue

Diffusion Tensor Imaging (DTI)

DTI is an extension of MRI that measures the **diffusion movement** of water molecules and thereby can **detect nerve fibers**.

Both methods are based on **nuclear magnetic resonance**.

Adapted from The Encyclopedia of Neuroscience



MRI head scan by Helmut Januschka licensed under CC BY-SA 3.0

Functional Neuroimaging

Electroencephalography (EEG) and Magnetoencephalography (MEG)

EEG and MEG enable the non-invasive monitoring of brain activity at high temporal resolution. While EEG measurements are based on measuring the **electrical potentials** resulting from neural activity via electrodes placed on the skull, MEG detects the corresponding **magnetic field**.

Positron Emission Tomography (PET) and Single-Photon Emission Computer Tomography (SPECT)

PET and SPECT are **nuclear imaging techniques** that require medication with radiotracers. Brain activity is monitored implicitly by observing the regional cerebral blood flow.

Functional MRI (fMRI)

fMRI is an extension of MRI that visualizes changes of the blood oxygen level caused by brain activity elicited through **external stimuli** or **cognitive processing** at high spatial resolution.

Adapted from The Encyclopedia of Neuroscience

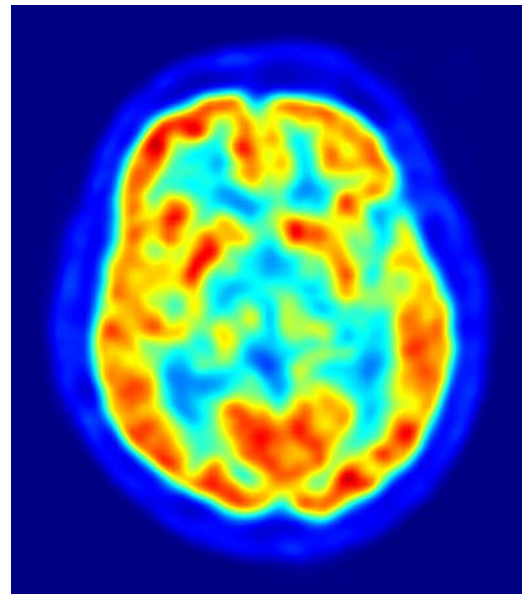
Examples

EEG



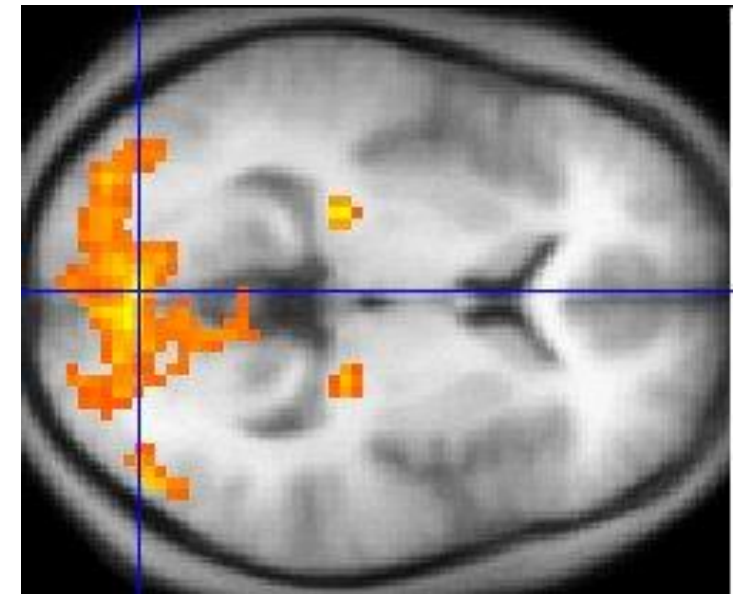
EEG plot by Der Lange licensed under CC BY-SA 2.0

PET



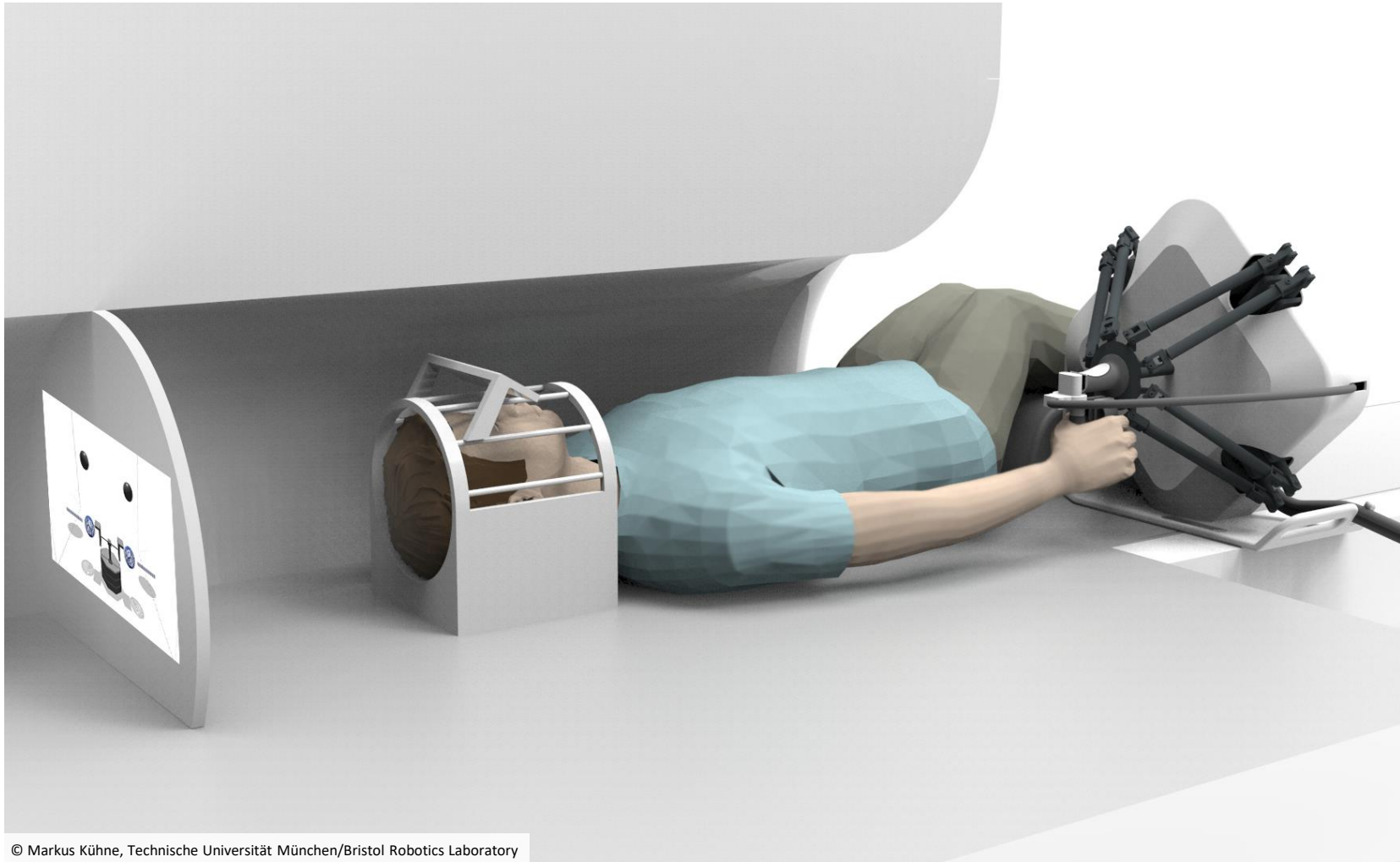
PET scan image by Jens Maus, Public Domain

fMRI



fMRI scan image by Washington Irving, Public Domain

Haptic Functional Brain Imaging

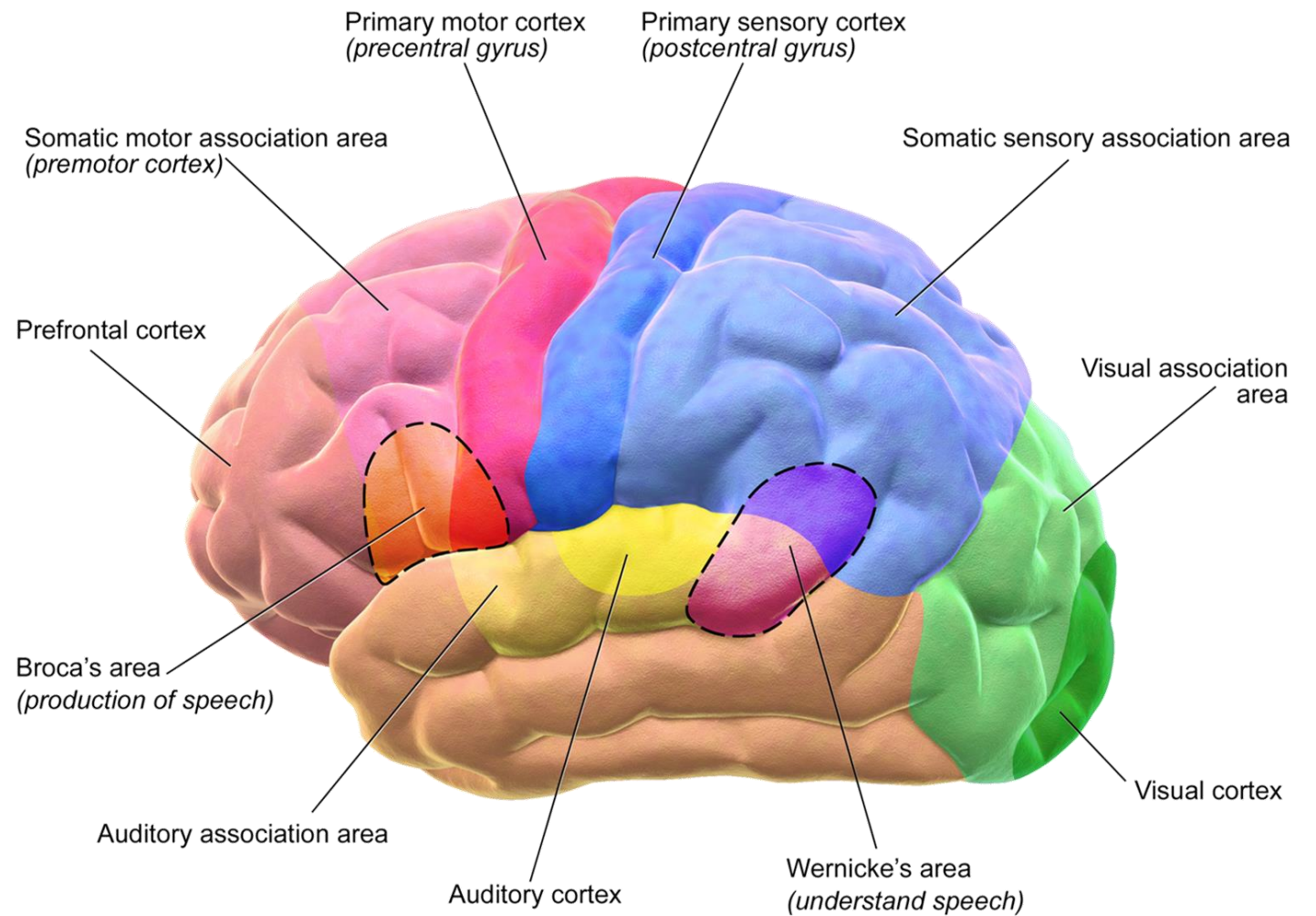


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Haptic fMRI augments fMRI with a **haptic input device**:

- The test person in the MRI scanner looks at **virtual objects** that are projected on a screen
- The haptic input device enables **interaction** with the virtual objects
- Brain activity (e.g. motor cortex) is **monitored** via fMRI

Motor and Sensory Regions of the Cerebral Cortex



Blausen.com staff (2014). "Medical gallery of Blausen Medical 2014". WikiJournal of Medicine 1 (2). DOI:10.15347/wjm/2014.010. ISSN 2002-4436. Licensed under CC BY 3.0
Cropped version of the original

Modeling Cognition – Important Prerequisites

Relevant Fields of Research

Engineering

Artificial Intelligence
Machine Learning
Computer Science
Robotics & Control
Chip Design
Subtopics of Mathematics

Natural Sciences

Neuroscience
Neurobiology
Developmental Sciences
Biomechanics
Neural Engineering
Neuroimaging

Humanities

Cognitive Psychology
Linguistics
Epistemology
Philosophy
Ethics

Cognitive Science

Biological Inspiration in Cognitive Modeling

Cognitive systems and functions can be modeled at different degrees of biological realism:

Computational Models

Purely computational models implement cognitive functions solely based on a functional view of the system without any reference to biology.

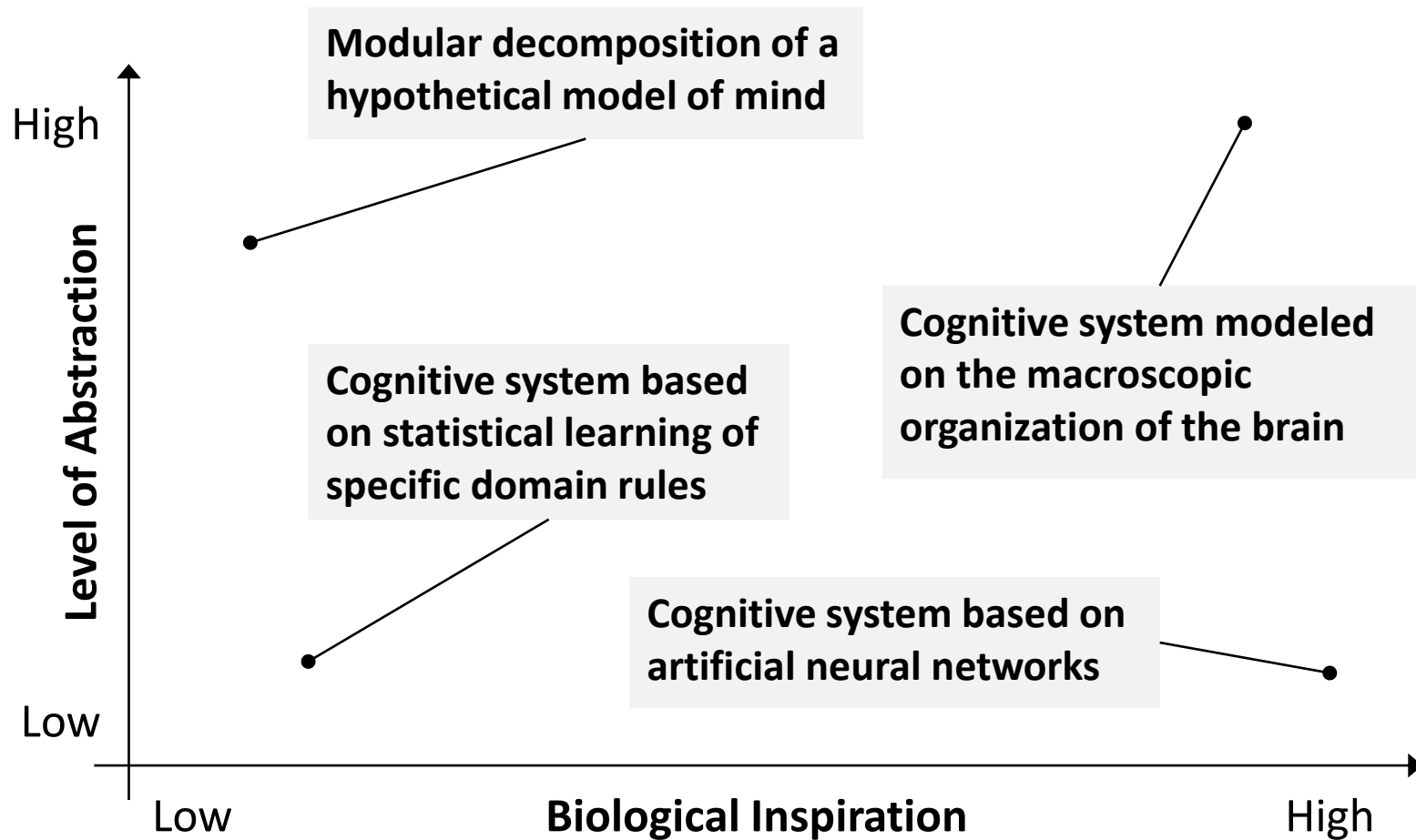
Bio-Inspired Models

Bio-inspired models implement cognitive functions by replicating known or hypothesized mechanisms of cognitive processing from biological organisms.

Hybrid Models

Hybrid models combine computational and bio-inspired modeling.

Computational vs. Bio-Inspired Modeling



Adapted from Vernon (2014)

The Ultimate-Proximate Distinction

Two Explanations for the Evolution of Cognitive Behavior

Cognition is specific to an **agent** and its **environment**. In the course of evolution, living creatures have developed cognitive skills that enable them to act and survive in their native environments. Every cognitive function known from biology therefore has two different explanations:

- The **ultimate** explanation: Why does a cognitive system exhibit a certain behavior?
- The **proximate** explanation: How is a certain behavior implemented in a cognitive system?

Consequence

To successfully develop a cognitive system requires that the implementation of cognitive functions with ultimate explanations match the system's specific working environment.

The Ultimate-Proximate Distinction

Example: Crying Baby

Ultimate Explanation

Crying elicits care and defense from caregivers. Babies that do not cry when in need have a lower chance of surviving

Proximate Explanation

Babies cry because of separation from their caregivers, cold, lack of food, and other internal mechanisms

Scott-Phillips, T. C., Dickins, T. E., & West, S. A. (2011). Evolutionary theory and the ultimate–proximate distinction in the human behavioral sciences. *Perspectives on Psychological Science*, 6(1), 38-47.



The Ultimate-Proximate Distinction – Definition

Definition 3.1: The Ultimate-Proximate Distinction

A specific behavior of a biological cognitive system has both an ultimate and a proximate explanation. Ultimate explanations are concerned with why a behavior exists, and proximate explanations are concerned with how it works. These two types of explanation are complementary and the distinction is critical to evolutionary explanation.

Adapted from Scott-Phillips et al. (2011)

Levels of Abstraction in Cognitive Modeling

Every cognitive system can be modeled at three hierarchical levels of abstraction. Depending on the modeling paradigm, the levels are **loosely or **tightly** coupled:**

Computational Theory

High-level description of the system that states the goal of the computation and the logic of the strategy by which it is carried out.

Representation and Algorithms

Realization of the computational theory in terms of input, output, representations and algorithms that perform the required transformations.

Hardware Implementation

Physical implementation of the representations and algorithms.

From Vernon (2014)